

Wazuh Agents: Architecture, Deployment, and Security Monitoring & Detection

A Research Paper in Endpoint Security

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This paper was prepared for educational purposes.

All technical references are drawn from official Wazuh documentation and peer-reviewed literature.

Abstract

As cyber threats grow in sophistication, organisations require scalable, cost-effective tools to monitor their endpoints and detect malicious activity in real time. Wazuh is an open-source, unified Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) platform that addresses these requirements through lightweight, multi-platform agents. This paper examines the Wazuh agent in depth, covering its modular architecture and core capabilities, its deployment and configuration across diverse operating systems, and its security monitoring and detection mechanisms — including file integrity monitoring, log analysis, malware detection, and security configuration assessment. The paper concludes by evaluating the strengths and limitations of Wazuh agents in enterprise and educational environments. The findings indicate that Wazuh agents provide comprehensive endpoint visibility with minimal performance overhead, making them a viable choice for organisations of all sizes.

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1 Introduction

The proliferation of sophisticated cyber attacks has made endpoint security a critical concern for organisations of all sizes. Traditional antivirus solutions are no longer sufficient on their own; modern security operations require continuous visibility into endpoint activity, rapid threat detection, and automated response capabilities. Security Information and Event Management (SIEM) platforms have emerged as a central pillar of modern security operations centres (SOCs), aggregating and correlating events from diverse sources to provide actionable intelligence.

Wazuh is a free and open-source security platform that unifies SIEM and XDR capabilities under a single framework. Originally forked from OSSEC, Wazuh has evolved into a mature, enterprise-grade solution trusted by thousands of organisations worldwide — from small businesses to large enterprises. At the core of the Wazuh architecture is the *Wazuh agent*: a lightweight, multi-platform software component installed on monitored endpoints that continuously collects security-relevant data and forwards it to a centralised server for analysis and alerting.

This paper focuses on two interconnected aspects of Wazuh agents: their **deployment and configuration**, and their **security monitoring and detection** capabilities. Section 2 introduces the overall Wazuh architecture. Section 3 covers the deployment process across different operating systems and environments. Section 4 analyses the key security monitoring and detection features of the agent. Section 5 discusses the strengths, limitations, and practical considerations. Section 6 concludes the paper.

2 Wazuh Architecture Overview

2.1 Platform Components

The Wazuh platform is composed of four major components that work in concert:

- **Wazuh Agent** — Installed on monitored endpoints; collects and forwards security data.

- **Wazuh Server** — Receives, decodes, and analyses data from agents; applies detection rules and generates alerts.
- **Wazuh Indexer** — Stores, indexes, and manages security data for fast querying (built on OpenSearch).
- **Wazuh Dashboard** — A web-based interface for visualising alerts, managing agents, and conducting threat hunting.

The server and its central components can be deployed as an all-in-one single-node installation or distributed across multiple nodes for high availability and scalability. Filebeat is used internally to forward analysed data from the Wazuh server to the Wazuh indexer over TLS-encrypted connections [1].

2.2 Agent–Server Communication

The Wazuh agent establishes a persistent, authenticated connection to the Wazuh server on TCP port 1514 (configurable). All communications between the agent and server are encrypted using AES (128-bit blocks, 256-bit keys) by default, with Blowfish encryption available as an alternative [1]. This ensures that sensitive telemetry is protected in transit. Data collected by agents — including logs, file changes, and system inventory — is forwarded in near real-time to the server for analysis.

2.3 Agentless Monitoring

For systems where installing the agent is not possible (e.g., network devices, firewalls, routers, and switches), Wazuh supports agentless monitoring. These devices can forward log data via Syslog, SSH, or external APIs, allowing the Wazuh server to perform analysis without a resident agent [1].

3 Deployment and Configuration

3.1 Supported Platforms

One of the key strengths of the Wazuh agent is its broad platform support. The agent runs on Linux, Windows, macOS, Solaris, AIX, and other operating systems, and can be deployed on laptops, desktops, servers, cloud instances, containers, and virtual machines [2]. It is designed with minimal resource consumption in mind, requiring on average only 35 MB of RAM [3].

3.2 Installation Methods

Wazuh provides several installation methods tailored to different environments:

Package Manager (Linux) On Debian-based systems, the official Wazuh APT repository is added and the agent installed via `apt-get`. On Red Hat-based systems, the Wazuh YUM repository is used instead.

Windows Installer A GUI or command-line installer (MSI package) is available for Windows endpoints. Administrator privileges are required. All agent files are stored by default in `C:\Program Files (x86)\ossec-agent` [4].

Automated Deployment For large-scale environments, Wazuh supports deployment via automation tools such as Puppet, Chef, SCCM, and Ansible [3]. The Ansible role `wazuh-agent` provides a ready-to-use playbook.

Windows Group Policy Objects (GPO) In Active Directory environments, the Wazuh agent MSI can be distributed silently across domain endpoints using GPO [5].

A minimal Linux deployment command (Debian/Ubuntu) takes the following form:

```
1 # Add the Wazuh GPG key and repository
2 curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | \
3   gpg --no-default-keyring \
4     --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg \
5     --import && \
```

```
6  chmod 644 /usr/share/keyrings/wazuh.gpg
7
8  echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] \
9  https://packages.wazuh.com/4.x/apt/ stable main" | \
10 tee -a /etc/apt/sources.list.d/wazuh.list
11
12 # Install the agent, specifying the manager IP
13 WAZUH_MANAGER="<MANAGER_IP>" apt-get install wazuh-agent
14
15 # Enable and start the agent service
16 systemctl daemon-reload
17 systemctl enable wazuh-agent
18 systemctl start wazuh-agent
```

Listing 1: Installing and enrolling the Wazuh agent on Linux (Debian/Ubuntu)

3.3 Enrollment and Configuration

After installation, the agent must be enrolled with the Wazuh manager. Wazuh supports several enrollment methods, including automatic enrollment via deployment variables (as shown above) and manual enrollment via the dashboard. Once enrolled, agents can be organised into *groups* — logical collections that share a common configuration. Groups allow administrators to apply tailored monitoring policies to endpoints grouped by role, location, or system type [9].

Agent configuration is managed through the `ossec.conf` XML file, which controls which modules are active, which directories are monitored for file integrity, which log files are collected, and what active response actions are permitted. Remote configuration management is possible through the Wazuh server, enabling centralised policy enforcement.

4 Security Monitoring and Detection

The Wazuh agent's modular architecture means each security function is handled by a dedicated module. These modules can be individually enabled, disabled, and configured

to suit the security requirements of the monitored endpoint.

4.1 Log Data Collection

The *Log Collector* module reads flat log files and Windows event logs, collecting operating system and application log messages. It supports XPath filters for Windows events and recognises multi-line log formats such as Linux Audit logs. Log entries can be enriched with additional metadata before being forwarded to the server [2].

The Wazuh server applies a two-stage log processing pipeline to each received event. In the first stage, *decoders* parse raw log messages into structured key–value pairs. In the second stage, *rules* are matched against the decoded data to identify security-relevant events and generate alerts [7].

4.2 File Integrity Monitoring (FIM)

The *File Integrity Monitoring* module watches the file system and reports when files are created, deleted, or modified. It tracks changes in file attributes, permissions, ownership, and content, and natively identifies which users and applications created or modified files [6]. FIM is commonly used for:

- Detecting unauthorised modifications to critical system files;
- Supporting change management processes;
- Demonstrating compliance with regulations such as PCI-DSS and HIPAA.

4.3 Security Configuration Assessment (SCA)

The *Security Configuration Assessment* module performs continuous configuration audits using checks based on CIS (Centre for Internet Security) benchmarks. Administrators can also define custom checks to enforce organisation-specific security policies. SCA results include recommendations for remediation, references to security standards, and mappings to regulatory compliance frameworks [2].

4.4 Malware Detection

Unlike traditional signature-based antivirus tools, the Wazuh malware detection module uses a *non-signature-based approach* to detect anomalies and potential rootkits. It monitors system calls and actively searches for hidden processes, hidden files, and hidden ports — indicators that malware may be attempting to conceal itself from standard detection methods [2].

4.5 Intrusion Detection and Rule-Based Analysis

The Wazuh server complements the agent’s capabilities with a *signature-based intrusion detection system (IDS)* that applies a regular expression engine to collected log data to identify indicators of compromise (IOCs) [6]. The platform includes thousands of built-in rules and decoders, and administrators can create custom rules to address organisation-specific threats. Rule severity levels are assigned to detected events, enabling analysts to prioritise their investigation.

Wazuh maps detected events to the MITRE ATT&CK framework, providing context about the adversary tactics and techniques involved and supporting structured threat hunting [8].

4.6 Vulnerability Detection

The *System Inventory* module periodically scans endpoints and collects data on installed applications, running processes, network interfaces, and open ports. This inventory is cross-referenced with the Wazuh Cyber Threat Intelligence (CTI) feed to identify software components with known vulnerabilities. Alerts include information on affected versions and recommended remediation steps.

4.7 Active Response

The *Active Response* module enables automated countermeasures when threats are detected. Default response actions include blocking a network connection, stopping a running

process, or deleting a malicious file. Administrators can also define custom response scripts tailored to their incident response plans [2]. Active response significantly reduces mean time to respond (MTTR) to incidents.

5 Discussion

5.1 Strengths

Wazuh agents present several significant advantages. First, they are **open source and free**, eliminating licensing costs that are often prohibitive for smaller organisations. Second, the modular architecture means administrators can enable only the capabilities they need, minimising unnecessary resource consumption. Third, broad platform support — spanning Linux, Windows, macOS, Solaris, and AIX — makes Wazuh suitable for heterogeneous environments. Fourth, the agent’s low average memory footprint (35 MB RAM) ensures it can be deployed on production endpoints without impacting performance [3].

The integration with third-party tools such as VirusTotal, TheHive, and PagerDuty, as well as support for the MITRE ATT&CK framework, increases the operational utility of the platform in mature security operations environments.

5.2 Limitations

Despite its strengths, Wazuh has several limitations worth acknowledging. The initial configuration and rule tuning can be complex and time-consuming, particularly for organisations without dedicated security staff. At scale, managing thousands of agents and processing large volumes of events requires careful infrastructure planning. Additionally, while machine learning integration has been explored in recent research — one study achieving 97.2% accuracy with Random Forest classifiers to reduce false positives [10] — such advanced capabilities are not yet natively built into the base Wazuh platform and require custom integration.

5.3 Practical Considerations

For deployment in large environments, organisations should consider automation tools such as Ansible or Puppet to streamline agent rollout. Agent grouping should be configured thoughtfully to ensure that monitoring policies are applied consistently. Regular updates to both the agent software and the rule sets are essential to maintain detection effectiveness against emerging threats.

6 Conclusion

The Wazuh agent is a powerful, flexible, and cost-effective component for endpoint security monitoring. Its modular design, broad platform support, and rich feature set — encompassing log collection, file integrity monitoring, malware detection, configuration assessment, vulnerability detection, and active response — make it a compelling option for organisations seeking comprehensive visibility into their IT environments. Deployment can be automated at scale using standard configuration management tools, and the agent’s low resource footprint ensures minimal impact on monitored systems.

While configuration complexity and rule management present challenges, the open-source nature of Wazuh means that a large community and extensive documentation are available to assist administrators. As cyber threats continue to evolve, Wazuh’s active development and integration with threat intelligence frameworks such as MITRE ATT&CK position it as a relevant and valuable tool for the foreseeable future.

Future research directions include deeper integration of machine learning-based anomaly detection into the Wazuh pipeline, as well as evaluation of Wazuh’s performance in cloud-native and container-centric architectures.

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